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标题	Photocatalyzed Aerobic Oxidation of Thiols to Disulfides Using Cu2O Polyhedra
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摘要（原文节选）

To further demonstrate semiconductor facet effect to photocatalytic organic transformations and address a lack of using simple polyhedral semiconductor particles for disulfide bond formation, Cu2O cubes, octahedra, and rhombic dodecahedra were used to photocatalyze aerobic oxidation of 4-methylbenzenethiol. After reaction condition optimization, Cu2O crystals and N,N,N′,N′-tetramethylethylenediamine (TMEDA) were added to 4-methylbenzenethiol in ethanol for thiol oxidation to 1,2-di-p-tolyldisulfane under 390 nm lig ⋯

数据表 · Table 1（5 行，含数值行 5 行）

Control Experiments for Thiol Oxidation to Disulfide

entry	Cu2O RDs	TMEDA	O2	light	yield (%)a
1	+	+	+	+	70
2	–	+	+	+	51
3	+	–	+	+	trace
4b	+	+	–	+	15
5	+	+	+	–	40

脚注（逐字）：aReaction conditions: Cu2O RDs (2.9 mg), 4-methylbenzenethiol (0.4 mmol), TMEDA (0.4 mmol), ethanol (3.0 mL), 390 nm LED (40 W). Triphenylmethane as an internal standard.bN2 atmosphere.

正文对应句（逐字）：Next, control experiments were performed (Table 1). In the absence of Cu2O crystals, the yield dropped to 51%. Without adding TMEDA, only a trace amount of product was formed, showing its importance to reactivity. Replacing oxygen with nitrogen resulted in minimal product formation. Without light ir

数据表 · Table 2（4 行，含数值行 4 行）

Photocatalytic Activity Comparison of Different Cu2O Crystals

entry	catalyst	amount (mg)	yield (%)a
1	commercial Cu2O	2.9	54
2	Cu2O cubes	1.6	57
3	Cu2O octahedra	1.1	63
4	Cu2O RDs	2.9	70

脚注（逐字）：aReaction conditions: 4-methylbenzenethiol (0.4 mmol), TMEDA (0.4 mmol), ethanol (3.0 mL). Triphenylmethane as the internal standard.

正文对应句（逐字）：Next, control experiments were performed (Table 1). In the absence of Cu2O crystals, the yield dropped to 51%. Without adding TMEDA, only a trace amount of product was formed, showing its importance to reactivity. Replacing oxygen with nitrogen resulted in minimal product formation. Without light ir

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